

CLABTOOLKIT: Reducing Repetitive Tasks in Neuroimaging Research 26.10.2026

Yasser Alemán-Gómez

Connectomics Lab, Lausanne University Hospital (CHUV)

Documentation: https://clabtoolkit.readthedocs.io
Source code: https://github.com/connectomicslab/clabtoolkit
Cite (*Under revision in Aperture Neuro*): https://arxiv.org/abs/2607.02638

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Motivation and objective

- **Motivation:**
  - **Routine** neuroimaging **operations** are **re-implemented** in **every project**. These ad hoc versions are rarely tested, which risks subtle **errors**, hurts **reproducibility** and raises the barrier for new users.
    - Operation examples[1]:
      - Remove, add and merge regions in a parcellation
      - Interpolate scalar maps, such as FA, along streamlines
      - Generate functional connectomes directly from a parcellation
      - Create unified morphometry tables, ready for statistical analyses
      - Generate slice images and a web page to check the quality of many images quickly

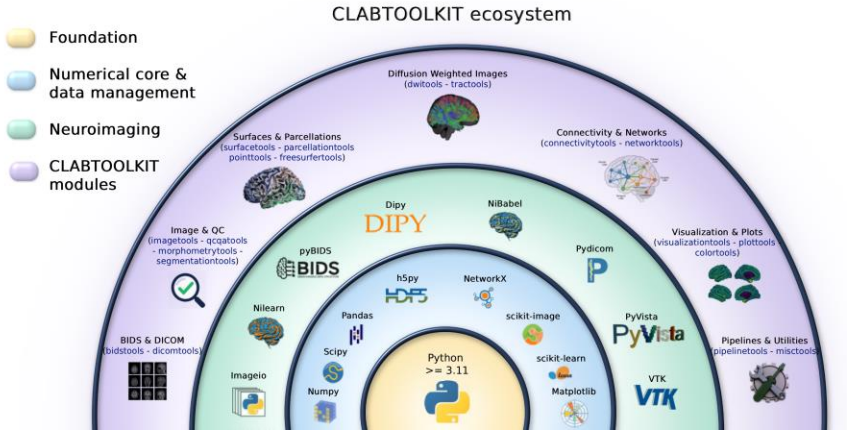
- **AIM:**
  - To **provide** a single, tested **layer** of shared **utilities** in Python that **removes repetitive**, error-prone **tasks**, building more robust, reproducible and FAIR workflows.

[1] https://clabtoolkit.readthedocs.io/en/latest/tasks.html

Ecosystem

CLABTOOLKIT is not a processing pipeline. It is a connector library.

Dependencies ecosystem



- **Main external and optional dependencies**
  - FreeSurfer
  - ANTs

The toolkit ships 5 JSON files

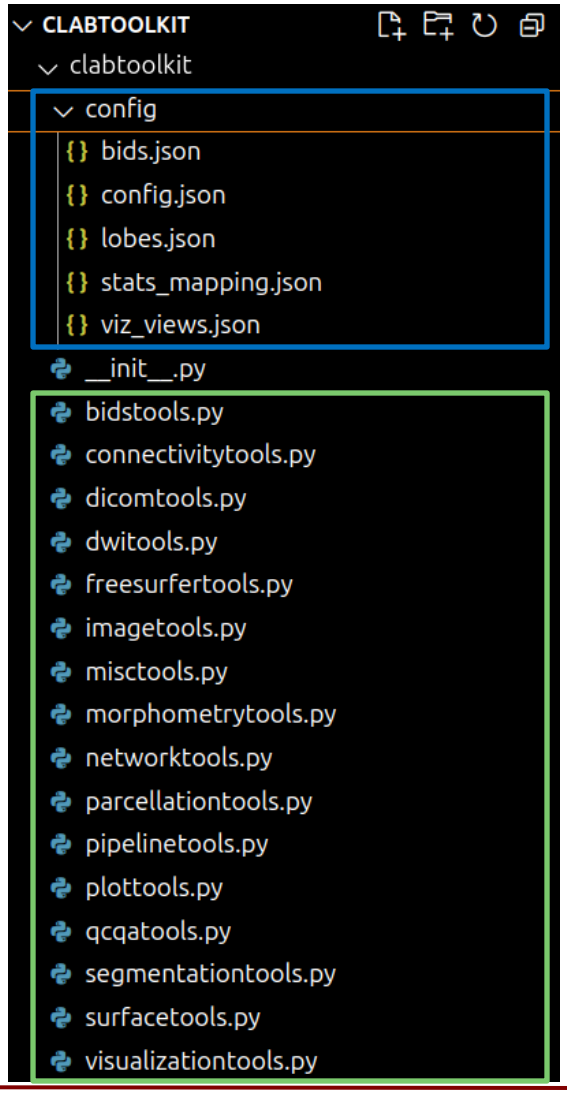
- **config.json:** *Units configuration*
  - Network metrics for binary or weighted graphs
  - Units
- **viz_views.json:** *Camera views, layouts and themes*
  - Windows size
  - Font colors
  - Layouts and themes
  - Views
- **lobes.json:** *Custom lobar grouping*
  - Lobar grouping
- **stats_mapping.json:** *Adapting the FreeSurfer parser*
- **bids.json:** *Handle BIDS entities*
  - Allowed raw and derivatives entities
  - Correct suffixes

CLABTOOLKIT Package architecture

- **Total Statistics:**
  - **19** modules
  - **27, 681** lines of code
  - **15** classes including **329** methods
  - **358** functions

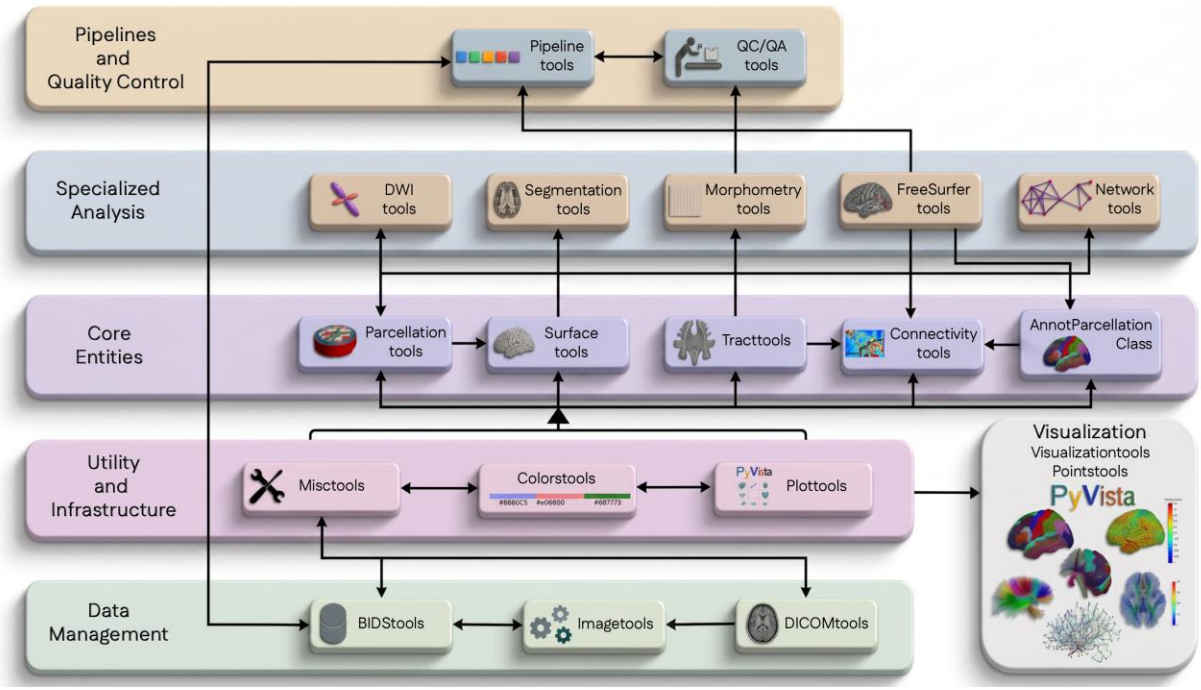
- **Key Modules:**
  - parcellationtools.py - **2828** lines
    (**2** classes, **66** methods)
  - visualizationtools.py - **2,548** lines
    (**1** class, **29** methods, **42** functions)
  - misctools.py - **4,947** lines
    (**1** classes, **1** method, **55** functions)

- **Principal Classes:**
  1. **BrainPlotter** (visualizationtools.py) - **29** methods
  2. **Surface** (surfacetools.py) - **47** methods
  3. **Parcellation** (parcellationtools.py) - **66** methods
  4. **Tractogram** (tracttools.py) - **29** methods



CLABTOOLKIT Package architecture

Architecture diagram of the CLABTOOLKIT modules

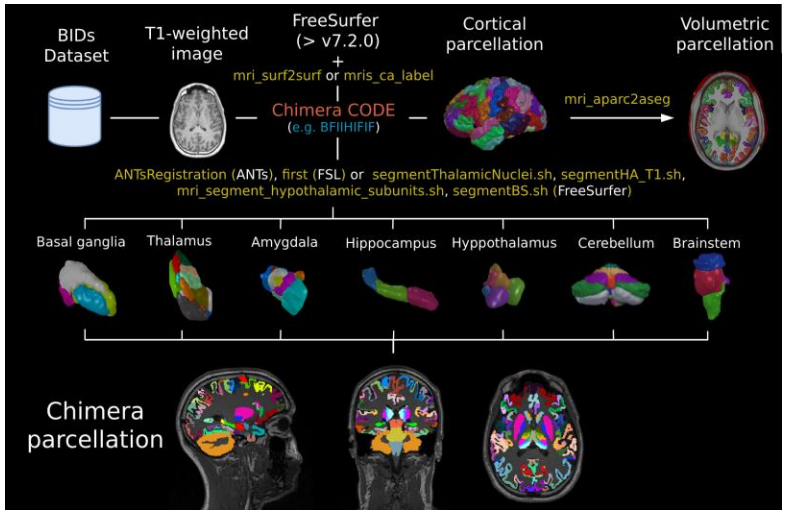


- **Architecture patterns:**
  - Specialized layered structure
  - Small set of core entities
  - Shared infrastructure
  - Configuration by JSON files
- **Strengths:**
  - Modular design
  - BIDS compliance
  - Reach visualization capabilities
- **Potencial issues:**
  - Circular dependencies (handled with conditional imports)
  - Large modules
- **Tests and continuous integration:**
  - Each module contains its own tests suite
  - ruff: linting
  - black: formatting
  - tox: Run the tests across Python versions locally

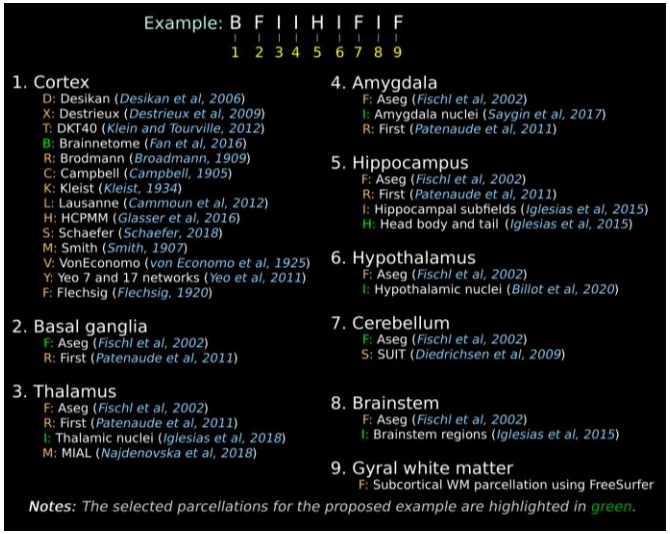
Applications (*parcellationtools*) → CHIMERA

https://github.com/connectomicslab/chimera
https://chimera-brainparcellation.readthedocs.io/en/latest/

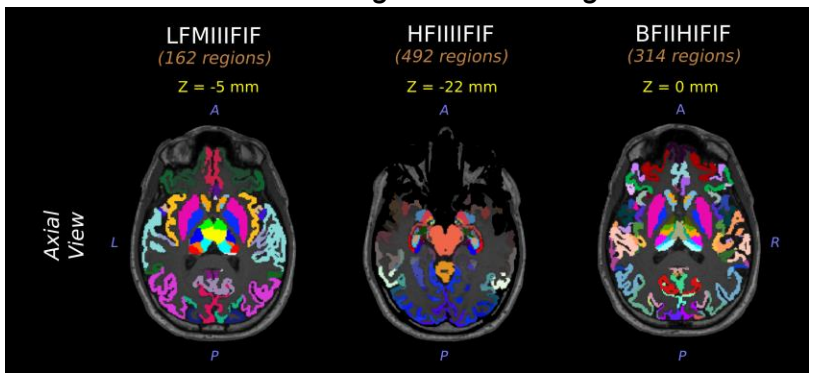
Methods: *Image processing workflow employed by CHIMERA*



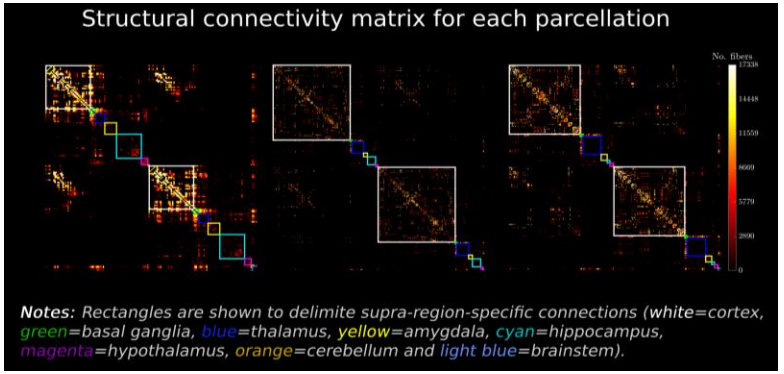
Available parcellations for each supra-region



Results: *Parcellations generated using CHIMERA*



Structural connectivity matrix for each parcellation



CLABTOOLKIT Demonstration

CLABTOOLKIT: Reducing Repetitive Tasks in Neuroimaging Research

(CBU Monday Methods Meeting 26.10.2026)

Table of Contents

Module: MiscTools (2 Classes, 22 Methods, 57 Functions)

- 1.1. Go to `build_indices` : Function to build the indexes from a range vector that can contain integers, tuples, lists or strings
- 1.2. Go to `generate_container_command` : This function generates the command to run a bash command inside a container

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Module: ColorsTools (2 Classes, 12 Methods, 25 Functions)

- 2.1. Go to `values2colors` : Map numerical values to colors using a specified colormap with optional inversions.
- 2.2. Go to `colortable_visualization` : Display a color table as an image

Module: ParcellationTools (2 Classes, 66 Methods, 0 Functions)

- 3.1. Go to `keep_by_name` : Filter the parcellation by region names
- 3.2. Go to `compute_centroids` : Compute centroids, number of voxels, and volume for each region
- 3.3. Go to `get_regionwise_timeseries` : Compute region-wise time series
- 3.4. Go to `compute_volume_table` : Calculate volumes for each region
- 3.5. Go to `surface_extraction` : Extract 3D surface meshes from parcellation regions

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Module: SurfaceTools (1 Classes, 47 Methods, 2 Functions)

- 4.1. Go to `load_scalar_maps` : Load morphometric data onto surface
- 4.2. Go to `load_annotation` : Load FreeSurfer annotation over the surface

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Questions?

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